

Program: Diploma in Microelectronics	
Course Code: 6492A	Course Title: Introduction to IoT
Semester: 6	Credits: 4
Course Category: Open Elective	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- Provide basic concepts of Internet of Things.

Course Pre-requisites:

Topic	Course code	Course name	Semester
Basics of computer		Introduction to IT systems	1
Fundamentals of Electronics		Fundamentals of Electrical and Electronics	2

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Explain the fundamental concepts of Internet of Things (IoT)	12	Understanding
CO2	Illustrate the IoT applications using Embedded Computing Board - Arduino	14	Applying
CO3	Identify the Python programming constructs for IoT	12	Applying
CO4	Illustrate the IoT applications using Embedded Computing Board - Raspberry PI	20	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	3						
CO3	3						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the fundamental concepts of Internet of Things (IoT)		
M1.01	Outline the key features of Internet of Things (IoT).	2	Understanding.
M1.02	Interpret IoT hardware, protocols and software using IoT Stack.	3	Understanding.
M1.03	Summarize different IoT enabling technologies.	3	Understanding.
M1.04	Classify IoT based on the complexity to build and operate.	3	Understanding.
M1.05	Explain challenges in IoT	1	Understanding
Contents: Introduction and Definition of IoT, Applications – Characteristics - Things in IoT - IoT Stack – IoT enabling Technologies – IoT challenges – IoT Levels			
CO2	Illustrate the IoT applications using Embedded Computing Board - Arduino		
M2.01	Illustrate the working of sensors and actuators	2	Understanding
M2.02	Outline the features of embedded computing boards - Arduino.	3	Understanding
M2.03	Illustrate the interfacing of sensors and actuators with Arduino	5	Applying
M 2.04	Illustrate the interaction with web page and cloud	4	Applying
	Series Test I	1	

Contents:

Sensors and Actuators - Role of Sensors and Actuators in IoT - Working of Sensors and Actuators - Examples.

Embedded Computing Boards - features and characteristics of Arduino/Node MCU, Interfacing with Arduino, Configure Arduino for IoT, Transferring data from Web and Cloud.

CO3	Identify the Python programming constructs for IoT		
M3.01	Explain the basic programming concepts of Python	2	Understanding.
M3.02	Illustrate the working of control structures in Python	4	Applying.
M3.03	Develop python programs to solve simple applications	4	Applying.
M3.04	Make use of Python packages for IoT	2	Applying.

Contents:

Python - variables and data types - executing a python program - control structures - python module - simple programs - python for IoT.

CO4	Illustrate the IoT applications using Embedded Computing Board - Raspberry PI		
M4.01	Outline the features of embedded computing boards - Raspberry PI	2	Understanding
M4.02	Illustrate the interfacing of basic sensors with embedded computing board - Raspberry PI	8	Understanding
M4.03	Case Study - the applications building with IoT - Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system etc	10	Understanding
	Series Test – II	1	

Contents:

Programming Raspberry Pi with python, Interfacing sensors and actuators with Raspberry PI.

Case Study - Applications building with IoT- Smart Perishable tracking/Smart transportation, Smart Healthcare, Smart Lavatory maintenance, Smart water through IoT, Smart warehouse monitoring, Smart Retail, Smart Driver assistance system.

Text / Reference

T/R	Book Title/Author
T1	RMD Sundaram Shriram K Vasudevan, Abhishek S Nagarajan, Internet of Things , Wiley Publications
R1	Vijay Madisetti, Arshdeep Bahga, Internet of Things: A Hands-On Approach , Orient Blackswan
R2	Raj Kamal, Internet of Things : Architecture And Design Principles , McGraw Hill Education
R3	Gary Smart, Practical Python Programming for IoT , Packt Publishing
R4	Marco Schwartz, Internet of Things with Arduino Cookbook , Packt Publishing

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/106/105/106105166/
2	https://www.raspberrypi.org/blog/getting-started-with-iot/
3	https://learn.adafruit.com/category/internet-of-things-iot
4	https://www.arduino.cc/en/IoT/HomePage
5	http://esp32.net/